

Practical No 6

```
#include <iostream>

#include <string>

using namespace std;

#define MAX 100

// ===== Array Queue for Ticket Reservation =====

class ArrayQueue {

    string queue[MAX];

    int front, rear;

public:

    ArrayQueue() {

        front = -1;

        rear = -1;

    }

    // Add a reservation request

    void enqueue(string request) {

        if (rear == MAX - 1) {

            cout << "Array Queue is full!" << endl;

            return;

        }

        if (front == -1) {
```

```
        front = 0;
    }
    rear++;
    queue[rear] = request;

    cout << "Request added: " << request << endl;
}

// Remove a reservation request
void dequeue() {
    if (front == -1 || front > rear) {
        cout << "Array Queue is empty!" << endl;
        return;
    }

    cout << "Request processed: " << queue[front] << endl;
    front++;

    if (front > rear) {
        front = -1;
        rear = -1;
    }
}

// Display queue
void display() {
```

```

    if (front == -1) {
        cout << "Array Queue is empty!" << endl;
        return;
    }
    cout << "Array Queue: ";
    for (int i = front; i <= rear; i++) {
        cout << queue[i] << " | ";
    }
    cout << endl;
}

};

// ===== Linked Queue for Ticket Reservation =====

struct Node {
    string request;
    Node* next;
};

class LinkedQueue {
    Node *front, *rear;

public:
    LinkedQueue() {
        front = NULL;
        rear = NULL;
    }

```

```
// Add a reservation request
```

```
void enqueue(string request) {
```

```
    Node* newNode = new Node;
```

```
    newNode->request = request;
```

```
    newNode->next = NULL;
```

```
    if (rear == NULL) {
```

```
        front = rear = newNode;
```

```
    } else {
```

```
        rear->next = newNode;
```

```
        rear = newNode;
```

```
    }
```

```
    cout << "Request added: " << request << endl;
```

```
}
```

```
// Remove a reservation request
```

```
void dequeue() {
```

```
    if (front == NULL) {
```

```
        cout << "Linked Queue is empty!" << endl;
```

```
        return;
```

```
    }
```

```
    Node* temp = front;
```

```
    cout << "Request processed: " << temp->request << endl;
```

```
    front = front->next;
```

```

        if (front == NULL)
            rear = NULL;

        delete temp;
    }

// Display queue
void display() {
    if (front == NULL) {
        cout << "Linked Queue is empty!" << endl;
        return;
    }
    Node* temp = front;
    cout << "Linked Queue: ";
    while (temp != NULL) {
        cout << temp->request << " | ";
        temp = temp->next;
    }
    cout << endl;
}

};

// ===== Main Function =====

int main() {
    ArrayQueue arrayQueue;
    LinkedQueue linkedQueue;

```

```
int choice;

string request;

do {

    cout << "\n===== TICKET RESERVATION SYSTEM =====" << endl;

    cout << "1. Add request to Array Queue" << endl;

    cout << "2. Process request from Array Queue" << endl;

    cout << "3. Display Array Queue" << endl;

    cout << "4. Add request to Linked List Queue" << endl;

    cout << "5. Process request from Linked List Queue" << endl;

    cout << "6. Display Linked List Queue" << endl;

    cout << "7. Exit" << endl;

    cout << "Enter your choice: ";

    cin >> choice;

    cin.ignore();

    switch (choice) {

        case 1:

            cout << "Enter ticket reservation request: ";

            getline(cin, request);

            arrayQueue.enqueue(request);

            break;

        case 2:

            arrayQueue.dequeue();

            break;

        case 3:
```

```
        arrayQueue.display();

        break;

    case 4:

        cout << "Enter ticket reservation request: ";

        getline(cin, request);

        linkedQueue.enqueue(request);

        break;

    case 5:

        linkedQueue.dequeue();

        break;

    case 6:

        linkedQueue.display();

        break;

    case 7:

        cout << "Exiting..." << endl;

        break;

    default:

        cout << "Invalid choice!" << endl;

    }

} while (choice!= 7);

return 0;

}
```

Output:

```
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===== TICKET RESERVATION SYSTEM =====
1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit
Enter your choice: 1
Enter ticket reservation request: Satara
Request added: Satara

===== TICKET RESERVATION SYSTEM =====
1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit
Enter your choice: 1
Enter ticket reservation request: Kolhapur
Request added: Kolhapur

===== TICKET RESERVATION SYSTEM =====
1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit
Enter your choice: 1
Enter ticket reservation request: Sangli
Request added: Sangli

===== TICKET RESERVATION SYSTEM =====
1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit
Enter your choice: 3
Array Queue: Satara | Kolhapur | Sangli |

===== TICKET RESERVATION SYSTEM =====
1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit
Enter your choice: 2
Request processed: Satara

===== TICKET RESERVATION SYSTEM =====
1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit
Enter your choice: 3
Array Queue: Kolhapur | Sangli |

===== TICKET RESERVATION SYSTEM =====
1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit
Enter your choice: 2
Request processed: Kolhapur

===== TICKET RESERVATION SYSTEM =====
1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit
Enter your choice: 3
Array Queue: Sangli |
```

===== TICKET RESERVATION SYSTEM =====

1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit

Enter your choice: 4

Enter ticket reservation request: Aurangabad

Request added: Aurangabad

===== TICKET RESERVATION SYSTEM =====

1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit

Enter your choice: 4

Enter ticket reservation request: Nashik

Request added: Nashik

===== TICKET RESERVATION SYSTEM =====

1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit

Enter your choice: 6

Linked Queue: Aurangabad | Nashik |

===== TICKET RESERVATION SYSTEM =====

1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit

Enter your choice: 5

Request processed: Aurangabad

===== TICKET RESERVATION SYSTEM =====

1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit

Enter your choice: 6

Linked Queue: Nashik |

===== TICKET RESERVATION SYSTEM =====

1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit

Enter your choice: 5

Request processed: Nashik

===== TICKET RESERVATION SYSTEM =====

1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit

Enter your choice: 6

Linked Queue is empty!

===== TICKET RESERVATION SYSTEM =====

1. Add request to Array Queue
2. Process request from Array Queue
3. Display Array Queue
4. Add request to Linked List Queue
5. Process request from Linked List Queue
6. Display Linked List Queue
7. Exit

Enter your choice: 7

Exiting...

Process exited after 207.5 seconds with return
Press any key to continue . . . |